Latent-class and mixture-model formulations for biomarker-treatment interaction in N-of-1 trials

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Contents

1 Abstract

Background. Aggregated N-of-1 clinical trials test biomarker-treatment interactions through a fixed continuous- heterogeneity regression in published practice. The psychometric literature has developed mature finite-mixture and latent-class machinery for representing discrete subpopulation structure that the biostatistical literature on N-of-1 trials has not engaged. The choice between continuous-heterogeneity and latent-class encodings of the biomarker-moderation question is a substantive scientific question rather than a parameterization convenience, but the field has not characterized when class-aware analyses become competitive with continuous regression at trial-relevant sample sizes.

Methods. We develop a five-component biomarker-moderation spectrum (covariance-only, mean-and-covariance, location-scale, heterogeneous random-effects, latent-class) bridging continuous-heterogeneity and latent-class formulations, and we catalogue the finite-mixture taxonomy (latent class analysis, growth mixture models, factor mixture models, regression mixtures) for application to within-subject longitudinal designs. We commit a heterogeneous-random-slopes generative data-generating process (DGP) and a class-aware analysis procedure (`lcmm::hlme` with gridsearch initialization), and run a 240-replicate-per-cell prototype simulation at $N = 70$, gating slope $\in \{0, 1.0, 1.5\}$, on a 4-core parallel implementation. Three test forms are recorded per replicate: the linear-mixed `bm:Dbc` Wald, the unconditional Wald on the lcmm gating coefficient, and a BIC-conditional gating procedure.

Results. In a 240-replicate-per-cell pilot we find that the unconditional Wald test on the latent-class gating coefficient inflates Type I error to 0.197 (95% CI $[0.145, 0.250]$) at the null, decisively above nominal 0.05 and consistent with the[1] and[2] overextraction predictions. The BIC-conditional gating procedure controls Type I (0.004 under the null) but is severely under-powered at trial-relevant N (rejection rate 0.025 at gating slope 1.0, 0.004 at 1.5). The linear-mixed `bm:Dbc` Wald test holds nominal Type I and reaches 0.542 power at gating slope 1.5, dominating both class-aware tests. The Monte Carlo standard error (MCSE) on power is approximately 0.030 across the grid. These pilot estimates precede the production five-study program described in the companion simulation plan, which has not yet been executed.

Conclusions. Within the 240-replicate pilot, at trial- relevant sample sizes ($N = 70$, eight measurement timepoints), neither the unconditional Wald-on-gating nor the BIC-conditional gating test is competitive with the linear-mixed baseline under the heterogeneous-random-slopes DGP. The latent-class encoding appears to require a regime of large class separation for class-aware tests to gain power; the pre-registered production program (five studies, $N \in \{35, 70, 100\}$, wider gating- slope grid) has not yet been executed and is required to map the regime where the BIC-conditional procedure becomes competitive.

2 Introduction

2.1 The aggregated N-of-1 trial and the biomarker-moderation question

The aggregated N-of-1 trial design is a class of clinical trials in which each enrolled participant cycles between active treatment and control across multiple within-person blocks, with within-person comparisons pooled across participants for population-level inference[3–6]. The design preserves the within-subject contrast that gives single-participant N-of-1 trials their inferential strength while accumulating the statistical efficiency of a multi-participant trial. For chronic conditions whose treatment response is reversible on a useful timescale, the design yields more information per participant per unit time than a parallel-group randomized trial of equivalent total sample size[7]. It has been deployed in chronic pain[8], inflammatory bowel disease, neuropsychiatric symptoms in post-traumatic stress disorder[9], and a growing portfolio of rare and orphan disease applications where parallel-group trials are infeasible[5,10].

Within this design class, a question of increasing scientific weight is whether treatment response is moderated by a measured pre-treatment biomarker, allowing clinicians to stratify treatment decisions at the point of care under the formal framework laid out in the PATH statement[11]. The corresponding statistical estimand is the biomarker-treatment interaction term in the within-participant outcome model. Power to detect this interaction depends on the magnitude of moderation, the precision of the within-participant contrasts, the trial design (open-label, crossover, hybrid), and any time-dependent leakage between treatment phases, conventionally termed ‘carryover’, that erodes the cleanness of the within-subject contrasts.

Recent methodological work has documented substantial power losses for biomarker-treatment interaction inference under realistic carryover, with the loss varying sharply by how the moderation effect is represented in the data-generating process[9]. The present paper takes up a question that the prior work raised but did not resolve: how should biomarker moderation be represented in the DGP itself? Two formal answers, drawn from a long history of work outside biostatistics, dominate the available options. We begin by laying out these two answers and the methodological tradition behind each, before turning to the specific consequences for N-of-1 trial design and analysis.

2.2 Two formal answers: continuous heterogeneity and latent-class structure

The first answer represents biomarker moderation as a graded continuous effect: every participant has some degree of treatment response, the response is a smooth (typically linear) function of the biomarker, and population heterogeneity is captured by a single multivariate distribution with an interaction parameter. This is the dominant tradition in clinical biostatistics, where heterogeneity of treatment effect is routinely modeled via random slopes on the treatment indicator and via interaction terms in fixed-effects regression. In its most economical form, the moderation enters the conditional mean of the outcome as $\beta_{bm:D} \cdot B \cdot D$, where B is the biomarker and D is the (possibly continuous-decay) treatment indicator.

The second answer represents biomarker moderation as a discrete subpopulation structure: participants belong to one of a small number of unobserved classes (most commonly 'responder' and 'non-responder'), the classes have qualitatively distinct treatment-response curves, and the biomarker predicts class membership without determining it. The corresponding generative form is a finite mixture in which class membership is a Bernoulli or multinomial draw with mixing probabilities that depend on the biomarker, and the within-class outcome model is class-specific. The marginal moderation observed in the data emerges from the mixture rather than from a smooth conditional-mean function.

These two answers correspond to genuinely different biological hypotheses. The continuous-heterogeneity view is consistent with a mechanism in which the biomarker reflects a graded physiological substrate (receptor density, enzymatic activity, baseline trait severity) that scales drug effect across the entire population. The latent-class view is consistent with a mechanism in which qualitatively distinct phenotypes exist (a metabolising versus non-metabolising allelic variant; a present versus absent target receptor) and the biomarker is an imperfect indicator of phenotype. The two views can produce identical marginal biomarker-outcome associations at a single timepoint and diverge sharply once the longitudinal structure of an N-of-1 trial is engaged, because carryover and within-person serial dependence interact differently with continuous and class-structured heterogeneity. Both traditions have long histories of formalization and identifiability analysis in their respective fields, and the corresponding tools are mature.

2.3 Related work: two methodological traditions

The continuous-heterogeneity and latent-class answers each arrive with a distinct history that this section summarizes; the full literature review, tracing both traditions decade by decade, is given in the companion supplement (`supplement.pdf`, Section 1).

The continuous-trait view originates with Spearman's two-factor theory of intelligence[12] and was elaborated through the modern factor-analytic tradition[13,14], in which between-person heterogeneity in a latent construct is by assumption continuous. The discrete-class view emerged from Lazarsfeld's latent structure analysis[15,16], placed on rigorous footing by Goodman[17] and unified in the finite-mixture monograph of McLachlan and Peel[18]. Bartholomew[19] framed the tension between the two views as a choice between 'graded' and 'unfolding' models that the data themselves rarely settle without strong prior assumptions.

A largely independent strand developed in econometrics and marketing. Heckman and Singer[20] showed that structural parameter estimates in duration models depend sensitively on the assumed heterogeneity distribution and proposed estimating that distribution as a discrete mixture, and Lindsay[21] placed the resulting theory of mixture likelihoods on a unified footing. In marketing science, DeSarbo and Cron[22], Wedel and DeSarbo[23], and Jedidi and colleagues[24] developed covariate-dependent gating functions for segment membership, a structure formalized independently in machine learning as 'mixtures of experts'[25]. This gating-function machinery is the exact structure the biomarker-moderation problem requires: whether a single observed covariate (the biomarker) predicts latent class membership, and whether the within-class treatment coefficient differs across classes.

By the late 1990s both strands converged on hybrid factor mixture models[2,26,27] and their longitudinal extension, the growth mixture model[28–32]. Bauer and Curran[1], Bauer[33], Lubke and Muthen[2], and Nylund and colleagues[34] subsequently documented overextraction hazards: growth and factor mixture models fitted to single-component, non-gaussian data can spuriously identify multiple latent classes, and reliably distinguishing genuine from spurious class structure typically requires several hundred to several thousand participants. This caution is central to the identifiability argument developed below.

Mixture methods entered longitudinal biostatistics through Verbeke and Lesaffre[35] and Zhang and Davidian[36], and through the `lcmm`[37] and `flexmix`[38,39] R packages used in the present paper's pilot. Despite this translation, mixture and latent-class methods have not, to the best of our knowledge, been systematically applied to aggregated N-of-1 trial designs: the published N-of-1 methods literature specifies biomarker moderation as a continuous interaction term throughout[5,6,9,10,40–42], with no parallel treatment of the latent-class formulation. This gap is what the present paper addresses.

2.4 Why this matters for N-of-1 designs specifically

Three features of the aggregated N-of-1 design make the continuous-versus-discrete moderation question particularly consequential for design and analysis decisions. First, the within-participant longitudinal structure of an N-of-1 trial provides repeated measurements at multiple drug states (on-drug, off-drug, in carryover). Under a continuous-heterogeneity DGP the moderation signal is carried by how the within-participant treatment contrast varies with the biomarker; under a latent-class DGP the signal is carried by class membership, which is fixed across all measurements within a participant and is therefore in principle recoverable from the full trajectory even when individual on-drug contrasts are noisy. The two formulations imply different efficient analysis strategies and different power profiles under carryover[9]. Second, N-of-1 trials are typically conducted at $N \in [30, 150]$, well below the sample sizes at which mixture-model identifiability has been validated in the psychometric literature, so the identifiability cautions documented by[1] and[2] apply with full force: even if a latent-class DGP is the biologically correct generative model, mixture analysis at trial-relevant sample sizes may be underpowered or unidentified, and a misspecified single-component analysis may be the more reliable inferential tool despite its bias. Resolving this trade-off is in part an empirical question the existing psychometric simulation literature has not addressed at N-of-1-relevant sample sizes. Third, carryover, which enters the analysis as a continuous decay in the effective drug indicator $D_{bc,it} = \exp(-\lambda t_{sd,it})$, interacts differently with the two encodings: it erodes the moderation signal directly in the continuous-heterogeneity formulation, where moderation enters as a product with $D_{bc,it}$, but erodes only the within-class drug contrast in the latent-class formulation, leaving class-membership information available if it can be recovered from the full trajectory. The two formulations therefore predict different power-loss profiles under carryover, a difference already documented to be empirically large in the companion carryover-sensitivity work.

2.5 Aims and structure of the paper

This paper has four aims: to formalize the latent-class and finite-mixture formulations of biomarker-treatment interaction in aggregated N-of-1 trials, with identifiability conditions specific to the trial-relevant sample-size regime; to characterize estimation strategies and the diagnostics required to distinguish genuine from spurious class structure at small N; to report a factorial simulation study benchmarking mixture-DGP and continuous-DGP power against the linear-mixed-model and class-aware analyses under realistic carryover; and to apply the developed methods to the prazosin-PTSD biomarker dataset of[9] in future work.

The remainder of the paper proceeds as follows. Notation fixes terminology; the latent-class generative form, its MVN second-moment approximation, and the regimes in which the approximation breaks down follow in three sections, then the implications for power under carryover and the identifiability constraints that bound mixture analysis at trial-relevant sample sizes. A finite-mixture taxonomy and a biomarker-moderation spectrum locate the relevant model families and DGP encodings in condensed form, with full development in the companion supplement. Implications for analysis and design, and software for mixture analyses, are then drawn together. The discussion addresses three open methodological questions and reports a 240-replicate pilot from the five-study simulation program described in the companion simulation plan, the empirical core of the present line of work.

3 Notation

Throughout, B_i denotes the pre-treatment biomarker value for participant i. BR_{it} denotes the biological-response component of outcome for participant i at time t, decomposed in the Hendrickson framework into three latent response factors: BR_{it} itself, a time-variant component TV_{it}, and a placebo-belief component PB_{it}. $D_{bc,it}$ denotes drug state at time t in the analysis-side continuous-decay form $D_{bc,it} = \exp(-\lambda t_{sd,it})$ during off-drug periods and $D_{bc,it} = 1$ during on-drug periods, with $t_{sd,it}$ the time since drug discontinuation and λ the decay rate. Following the convention used throughout this series, D_{it} is reserved for the binary on-drug state and $D_{bc,it}$ for the continuous exposure-decayed indicator used here; the latter is the `Dbc` regressor of the analysis model. Z_i denotes an unobserved class label. The half-life of carryover is $t_{1/2} = \ln 2/\lambda$ and is allowed to differ between the DGP and the analysis model.

The principal psychometric and econometric terms used below are:

- **Latent class.** An unobserved categorical class, equivalent in trial vocabulary to an unobserved responder/non-responder class.
- **Latent factor.** An unobserved continuous trait governing covariation among observed variables, equivalent to a frailty (the survival-analysis term for an unobserved multiplicative random effect) or to a random intercept in a linear mixed model.
- **Class-conditional distribution f_z.** The outcome distribution within a single latent class.
- **Mixing proportion π.** The prevalence of a given latent class in the population.
- **Measurement invariance.** Whether a model applies identically across subpopulations.
- **Growth mixture model (GMM).** A longitudinal mixed model in which the population of random effects is itself a finite mixture.
- **Factor mixture model (FMM).** A hybrid model with discrete classes at the upper level and within-class continuous heterogeneity at the lower level.
- **Regression mixture / mixture of experts.** A finite mixture in which the gating function (class-membership probability) depends on covariates and within-class regressions are class-specific.
- **EM algorithm.** Expectation-maximization (EM) applied to recover unobserved class labels jointly with class-conditional model parameters.

4 Faithful latent-class data-generating processes

If a candidate predictive biomarker indexes an unobserved responder/non-responder dichotomy, the mechanistically faithful data-generating process is a finite mixture:

$$Z_i \mid B_i \sim \text{Bernoulli}(\pi(B_i)), \qquad BR_{it} \mid Z_i = z \sim f_z(t, D_{bc,it}),$$

where Z_i is the latent class indicator, $\pi(B_i) = \Pr(Z_i = 1 \mid B_i)$ is the class-membership probability (typically a logistic function of the biomarker), and the class-conditional response distributions f_0 and f_1 differ in their drug-response parameters: peak response, onset rate, or both. Within-class temporal structure follows the same modGompertz form used in single-class formulations, with class-specific parameters substituted.

Two consequences of this DGP form are central to what follows. First, the class label Z_i is fixed across all measurements within a participant, invariant to drug state and to time. Second, the class-conditional mean trajectory does depend on $D_{bc,it}$: under a full carryover off-drug period ($D_{bc,it} \to 0$) the two class-conditional means converge if the responder/non-responder distinction is purely in drug response. Class-label invariance therefore preserves participant-level information about the moderation signal that an analysis with access to Z_i could exploit, even where instantaneous between-class mean separation in BR is fully attenuated. The MVN-covariance encoding, in contrast, has no such latent-label information to exploit: the covariance is the entire signal, and its erosion under carryover is direct.

5 MVN approximation to mixture-induced moderation

A two-component mixture with class-dependent means and common within-class covariance produces a marginal joint distribution of (B_i, BR_{it}) whose first two moments are reproduced, to leading order, by a single multivariate normal with $\text{Cor}(B, BR) = c_{bm}$. Under within-class independence between B_i and BR_{it},

$$c_{bm}^2 \;=\; \frac{\pi(1-\pi)(\mu_B^1 - \mu_B^0)^2}{\text{Var}(B)} \;\cdot\; \frac{\pi(1-\pi)(\mu_{BR}^1 - \mu_{BR}^0)^2}{\text{Var}(BR)},$$

so that c_{bm}^2 is the product of the between-class variance fractions in B and in BR. The marginal correlation that an MVN can reproduce is therefore bounded by the geometric mean of the two intra-class correlations; if either class separation is small, the achievable marginal c_{bm} is small even when class structure is otherwise well-defined.

Under mild regularity conditions (adequate class overlap and mixing proportions π bounded away from 0 and 1), the mixture's second-moment structure is well-approximated by a single MVN with differential correlation across drug states: elevated correlation where class-conditional means separate (on-drug) and shrinkage towards zero where they coincide (off-drug after carryover has decayed). For power calculations driven by t-tests on $\beta_{bm:D}$ in a linear mixed-effects model, this approximation is adequate within the regularity conditions and captures the covariance consequences of latent-class structure without committing to a discrete generative form.

6 Failure regimes of the MVN approximation

The MVN approximation is misleading in three identifiable regimes, each with a distinct empirical signature. Under **bimodal responses**, when f_0 and f_1 are well-separated, the marginal distribution of BR is genuinely bimodal, no MVN matches its tails, and power under the true mixture DGP will exceed the MVN-LMM prediction because a well-specified mixture analysis can exploit the bimodality; the pattern occurs where a subpopulation genuinely does not respond, as in certain targeted oncology indications. Under a **strong class-membership gradient**, when $\pi(B_i)$ is steep and approaches a step function at a biomarker threshold, the biomarker behaves nearly deterministically as a class label and a thresholded mean-moderation specification becomes the better approximation, corresponding clinically to a predictive biomarker with near-perfect sensitivity and specificity. Under **class-varying covariance**, if autocorrelation, residual variance, or placebo-response parameters differ between classes, no single-component MVN can represent the generative structure and analysis models assuming constant covariance are misspecified regardless of whether the biomarker interaction is represented in the mean or the covariance; this is the latent-class analogue of heteroscedastic residual variance addressed in standard mixed-effects practice with `varIdent` or `weights` specifications, except that the stratifying variable is unobserved.

7 Implications for power under carryover

The substantial covariance-only power loss under carryover (40–60% across designs in[9]) reflects erosion of a second-moment signal. A mixture-model analysis fitted to mixture-DGP data can in principle recover additional power by estimating class membership directly, because the class label is carryover-invariant. The gap between the covariance-only carryover-sensitive power estimate and the mixture-faithful power estimate upper-bounds what a mixture analysis could recover; the bound is asymptotic, and at finite N finite-sample bias can drive the realized gap below the bound or, in adverse identifiability regimes, reverse its sign. A focused factorial simulation isolating second-moment approximation error from analysis-model mis-specification is described in the companion simulation plan (Study 1: mixture-vs-MVN factorial), crossing mixture and MVN DGPs with linear-mixed and joint latent-class-mixed analyses; the mixture-DGP/MVN-analysis versus mixture-DGP/mixture-analysis contrast quantifies power recoverable by a faithful analysis model, and a null check on the MVN-DGP/mixture-analysis cell verifies the mixture model does not extract spurious power where no latent-class structure exists.

8 Identifiability at trial-relevant sample sizes

Mixture modeling is attractive on faithfulness grounds but carries identifiability costs that are acute in N-of-1 settings. The EM algorithm for finite mixtures of mixed-effects models requires either informative priors on class-membership probabilities or strong separation between class-conditional response curves, and the standard diagnostic for mixture identifiability, the entropy of posterior class-membership probabilities, requires substantial sample sizes (N several hundred to several thousand) to stabilize at values supporting confident class extraction in published psychometric applications[1,33,34]. Aggregated N-of-1 trials of the size contemplated in current biomarker applications, $N \in [30, 150]$, are well below this range, so mixture analysis at trial-relevant sample sizes may be underpowered or unidentified even when the underlying DGP is genuinely mixture-structured. The question is therefore not whether mixture formulations are more faithful, but whether the fidelity gain is recoverable at realistic sample sizes, or whether analysis-strategy mitigations within standard mixed-effects practice (restricted analysis, weighting, within-subject contrasts; see[9]), which have better-understood inferential properties at the relevant sample sizes, deliver a better power-per-complexity trade. Resolving this empirically requires the simulation program described in the companion plan; Study 5 addresses small-N identifiability and Type I error directly.

9 Finite-mixture taxonomy

Four families of finite mixture model from the psychometric and econometric literatures bear directly on the biomarker-treatment interaction problem. Table 1 summarizes each family's structure, origin, and relevance to the N-of-1 setting; the full development of each, including the reasoning behind the package recommendations, is given in the companion supplement (Section 2).

Table 1: Finite-mixture model families relevant to the biomarker-moderation problem.

Family	Structure	Origin	Relevance to N-of-1 trials
Latent class / latent profile analysis (LCA/LPA)	Cross-sectional mixture; class-specific marginal distributions	[16];[17];[18]	Baseline responder-class discovery from biomarker panels; `poLCA`

Family	Structure	Origin	Relevance to N-of-1 trials
Growth mixture model (GMM)	Longitudinal mixed model with a class-mixture random-effects population; class label is drug-state-invariant	[28;31]	Closest analogue of the faithful latent-class DGP in Section 3; overextraction cautions[1] apply with full force at N-of-1 sample sizes
Factor mixture model (FMM)	Discrete classes at the upper level, continuous within-class factor structure at the lower level	[26;27;2]	Biomarker as both a noisy class indicator and a continuous within-class modulator; higher estimation cost
Regression mixture / mixture of experts	Class-specific regression surfaces with a covariate-dependent gating function	[22;25]	Biomarker governs both class probability and within-class drug-effect magnitude; `flexmix`

GMM is the closest psychometric analogue of the faithful latent-class generative form developed in Section 3: each participant belongs to a latent class with its own drug-response trajectory, and a GMM that uses the biomarker as a class-membership predictor is essentially that generative model written in psychometric vocabulary. This is why the overextraction cautions from the GMM literature[1,33] carry directly into the identifiability discussion below.

10 The biomarker-moderation spectrum

The covariance-only formulation of[9] specifies a biomarker-dependent covariance structure with population means held constant across biomarker values. Locating this formulation within the broader spectrum of biomarker-moderated DGPs is useful because the moment-by-moment organization of the spectrum (biomarker moderating the mean, the covariance, both, or higher moments) clarifies which features of the carryover-induced power loss are properties of the specification itself rather than of the underlying scientific question.

Table 2 organizes the five points on the spectrum moment by moment; the full development of each, including the measurement-invariance parallel for the covariance-only case and the GAMLSS distributional-regression formalism, is given in the companion supplement (Section 3).

Table 2: The biomarker-moderation spectrum, organized by which moment of the response distribution carries the biomarker signal.

Point on spectrum	What varies by class/biomarker	Behaviour under carryover
Covariance-only[9]	$\mathrm{Cov}(B, BR)$ only; means constant	Signal is entirely in the eroded channel; power loss is largest
Combined mean and covariance[27]	Both $E[BR \mid B]$ and $\mathrm{Var}[BR \mid B]$	Mean channel survives carryover; power loss intermediate
Location-scale (GAMLSS)[43,44]	Location and scale of the marginal distribution, no discrete classes	Analogue of the combined case without committing to a class count
Heterogeneous random-effects[35,37]	Class-specific random-intercept/slope distributions	Class label is carryover-invariant; used for the Study 5 pilot below
Pure mean moderation (Architecture A)	$E[BR \mid B]$ only; covariance constant	Signal survives carryover essentially undiminished

The covariance-only specification of[9] is the most restrictive point on this spectrum: it holds $E[BR_{it}]$ fixed across biomarker strata and places the entire interaction in $\mathrm{Cov}(B_i, BR_{it} \mid D_{bc,it})$, which is an unusual restriction within the psychometric mixture tradition (most standard FMM and GMM parameterizations permit class-specific means at a minimum). Its appeal is tractability for power simulation under multivariate normality; its cost is that it does not correspond cleanly to a natural generative mechanism, and, because the entire signal sits in a channel that carryover erodes by construction, a sensitivity analysis based solely on this point risks overstating the carryover-induced power loss a biomarker-guided trial should actually expect. The heterogeneous random-effects row is the point used for the Study 5 pilot reported below, because it preserves compatibility with the existing `nlme::lme` analysis machinery while admitting a biologically faithful latent-class generative structure at the DGP level; the `lcmm` package[37] implements this family directly.

11 Implications for analysis and design

Three implications follow from locating the covariance-only encoding within the broader spectrum. First, the covariance-only parameterization is the most restrictive operational case of a much richer family: the substantial power loss under carryover documented in[9] is a specific property of this restrictive case, because it places the entire biomarker-outcome signal in a channel carryover erodes by construction, and need not generalize to the mean-plus-covariance or heterogeneous-random-effects variants standard in psychometric and biostatistical practice. A complete sensitivity analysis should report power under at least three points on the spectrum: pure mean moderation, pure covariance moderation, and a combined specification. Second, the identifiability concerns documented in the GMM and FMM literatures[1,2,33,34] bound what can be asked of mixture-based analysis models at N-of-1 sample sizes: published FMM applications typically involve several hundred to several thousand participants, an order of magnitude above biomarker-trial applications with $N \in [30, 150]$, so analysis plans built around fitting `lcmm` or `flexmix` directly to trial data should not be relied on as the primary analysis, though mixture analysis retains value as a sensitivity check and as a DGP for power simulation. Third, if N-of-1 simulation infrastructure pursues a richer DGP in subsequent work, the natural target is the heterogeneous random-slopes form rather than a full FMM, because it preserves the analysis model's compatibility with the existing `nlme::lme` machinery while admitting a biologically faithful latent-class generative structure at the DGP level, preserving the audit chain of the existing framework and permitting direct comparison against the covariance-only baseline under matched effect-size tuning.

12 Software for mixture analyses

R packages relevant to estimating the models surveyed above, in approximate order of fit to the longitudinal aggregated N-of-1 use case: `lcmm`[37], the closest available tooling for heterogeneous random-effects specification, with syntax closely related to `nlme::lme`; `flexmix`[38,39], a more general regression-mixture framework requiring more configuration; `mclust`[45], for cross-sectional latent-profile analyses of baseline biomarker panels; `gamlss`[43], for distributional regression; `poLCA`[46], for categorical-indicator LCA; and `OpenMx` and `lavaan` with mixture extensions, for FMM-style structural-equation specifications at higher configuration cost. The reference implementation outside R is Mplus[47], the standard environment for GMM, FMM, and related mixture-SEM work, and the environment used by much of the identifiability and overextraction literature cited above[1,2,34].

13 Discussion

The biomarker-treatment moderation problem in aggregated N-of-1 trials sits at the intersection of two methodological traditions that have, until now, only weakly engaged with each other: a biostatistical tradition with efficient within-participant estimators but a continuous-interaction-only specification of moderation, and a psychometric tradition with mature finite-mixture machinery applied primarily at sample sizes well above the N-of-1 range. The argument of this paper is that the choice between continuous-heterogeneity and latent-class encodings is a substantive scientific question, not a parameterization convenience, opening both a sensitivity-analysis program (the moderation spectrum surveyed above) and an analysis-strategy program (class-aware mixture analyses where identifiability permits, mitigations where it does not).

Two limitations bound what the paper currently provides. First, the 240-replicate pilot reported below is a single cell of the five-study production factorial (mixture-vs-MVN, the mean-covariance-combined triptych, heterogeneous random slopes, the three-regime breakdown grid, and the small-N identifiability pre-flight) described in the companion simulation plan; the production run at 1,000+ replicates per cell has not yet been executed. Second, the application to real biomarker-guided trial data has not been carried out; the prazosin-PTSD analysis of[9] provides the most immediate target for re-analysis under at least three biomarker-moderation encodings once the production run completes.

Three open methodological questions remain, to be addressed by the production program. The first is whether class-aware analyses are usable as primary analyses at any realistic N-of-1 sample size, or only as sensitivity checks; the pilot evidence below suggests the latter, and only the wider production grid can settle it definitively. The second is how analysis half-life should be specified when the DGP is a mixture rather than a single MVN, since the continuous-decay drug indicator $D_{bc,it} = \exp(-\lambda t_{sd,it})$ has a clean interpretation under a single-component generative model but a less obvious one when class-conditional carryover dynamics differ between classes. The third is whether the trial design itself should respond to the moderation-encoding question: if a latent-class generative form is biologically more plausible than a continuous one, designs that maximize the per-participant contrast and minimize carryover may gain power that the standard hybrid design surrenders.

Two secondary theoretical points are developed fully in the companion supplement (Section 5). First, the analysis model used throughout this program is a four-term linear mixed model rather than a component-matched non-linear model mirroring the DGP's biological-response, placebo-belief, and time-variant decomposition; the supplement gives four reasons this simpler analysis is the right choice at trial-relevant N. Second, the latent-class DGP specified for Study 1 is a natural mechanism for generating apparent subadditivity in a downstream single-component analysis, even absent any direct mechanistic interaction; the Study 1 pre-registration includes an axis designed to distinguish this emergent mechanism from genuine mechanistic interaction empirically.

13.1 Progress to date

The five-study production program is pre-registered under the ADEMP framework of Morris, White, and Crowther[48] at `analysis/scripts/latent-class-mixture-application` `00-ademp-pre-registration.md` (approximately 2,300 cells across the program, seeded random-number control, an effect-size tuning substudy, and a deviation-log specification). Phase 1 has reached the infrastructure-and-pilot milestone: a class-aware analysis wrapper around `lcmm::hlme` with the same input-output contract as the existing `lme_analysis()`, and a heterogeneous-random-slopes DGP wrapper around `generateData()` that adds a Bernoulli class-membership channel with logistic gating in the biomarker. An earlier 100-replicate pre-flight pilot under the null DGP compared six candidate class-aware test forms and found that an unconditional Wald test on the `lcmm` gating coefficient inflates Type I error to five times nominal (0.250 against 0.05), while a BIC-conditional gating procedure (fit `ng=1` and `ng=2`, test the gating coefficient only if BIC prefers `ng=2`) controls it. That pre-flight comparison, reported in full in the companion supplement (Section 6), fixed the BIC-conditional procedure as the primary class-aware moderation test carried into the production-scale pilot below. Extrapolated to the pre-registered cell counts, the class-aware arm of Study 1 alone is approximately 340 hours of serial CPU time (22–44 hours parallelised across 8–16 cores); the full five-study program is feasible on a single workstation over a week of background compute.

13.2 Pilot results (Study 5, 240 replicates)

The Study 5 prototype was rerun at 240 replicates per cell on 4-core `parallel::mclapply` with three gating-slope levels (gs $\in \{0, 1.0, 1.5\}$) and three test forms (lme `bm:Dbc` Wald, `lcmm` gating Wald unconditional, BIC-conditional gating). Wall time was 558 s; convergence was 92.9% under the null and 98.3–98.8% at non-null cells. Per-replicate output is at `analysis/data/quick-sim/03-latent-replicates.rds`; Morris ADEMP summary at `03-latent-summary.txt`. MCSE at the working power near 0.5 is approximately 0.030.

The unconditional Wald-on-gating Type I error of 0.197 (95% binomial CI $[0.145, 0.250]$) is now decisively distinguishable from the nominal 0.05 (more than five MCSE of separation) and is statistically consistent with the 100-replicate pilot's 0.250 point estimate. Power at gs = 1.0 (0.304) and gs = 1.5 (0.360) sits below the lme baseline (0.325 and 0.542 respectively), so the unconditional Wald buys nothing in exchange for the Type I inflation. The BIC-conditional procedure does control Type I (0.004 under the null, with upper CI near 0.013), but its power is 0.025 at gs = 1.0 and 0.004 at gs = 1.5: under the heterogeneous-random-slopes DGP at $N = 70$, the BIC step selects `ng=1` in essentially every replicate and the conditional procedure approximates a never-reject rule. The non-monotonicity (gs = 1.5 below gs = 1.0) suggests that as the heterogeneous-slopes signal strengthens, a single-class lme captures it well enough that BIC continues to favor `ng=1` and the conditional procedure does not fire.

The lme `bm:Dbc` test holds nominal Type I (0 of 240 under the null) and reaches 0.542 power at gs = 1.5, dominating both `lcmm` tests at this sample size and DGP regime. We therefore conclude, on the evidence of this pilot, that at trial-relevant N and under the heterogeneous-random-slopes generating mechanism examined here, neither the unconditional Wald-on-gating nor the BIC-conditional gating test is competitive with the linear-mixed baseline: the unconditional test is severely anti-conservative, and the BIC-conditional test is severely under-powered. A canonical run at 1,000 replicates per cell with a wider gating-slope grid (and at $N \in \{35, 70, 100\}$) is required to map the regime in which the BIC-conditional procedure does become competitive, plausibly at large class separations rather than under heterogeneous random slopes.

14 Conflict of interest

The authors declare no conflicts of interest.

15 Funding

This work received no specific funding.

16 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte

Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

17 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash in `Dockerfile`; package set in `renv.lock`). Principal packages are `nlme` (`corCAR1` linear-mixed analysis), `parallel` (4-core `mclapply`), `lcmm` (class-aware analysis, `gridsearch` initialization), and `rmarkdown`/`bookdown` for the manuscript build. Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop, with per-replicate seeds derived by `+ rep_idx` under `parallel::mc.set.seed`.

Per-replicate outputs from the 240-replicate Study 5 pilot are checkpointed at `analysis/data/quick-sim/03-latent-replicates.rds`, with Morris ADEMP cell-level summaries at `03-latent-summary.txt` and the effect-size calibration table at `03-calibration-table.rds`. Driver scripts are under `analysis/scripts/latent-class-mixture-application/`, including the pilot driver `03-study5-pilot.R` (executed) and the production driver `10-study5-production.R` (committed; not yet run). The pre-registered ADEMP plan is at `00-ademp-pre-registration.md` in the same directory. The manuscript source, bibliography, and CSL file are at `analysis/report/03-latent-class-mixture-application/`.

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